

7. Carwyn and Anwen are year 10 students who are concerned that they are overweight. They think that their health may be affected in later life. They both agree to help each other follow a balanced diet.

They research the topic on the Internet and find the following information:

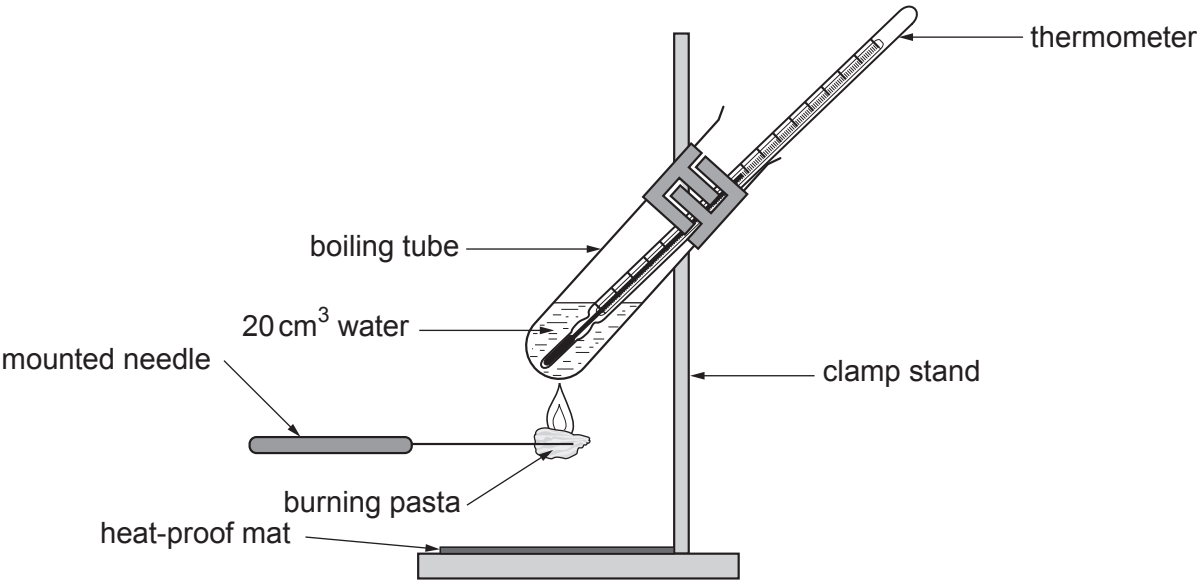
As part of a healthy balanced diet, it is recommended that people of their age should be eating the following every day:

Energy	8400 kJ
Total fat	70 g
Starchy carbohydrate (e.g. bread, pasta, rice)	260 g
Total sugar (from milk, fruit and added sugar)	90 g
Protein	50 g
Salt	6 g

Carwyn does not want more than one third of his recommended energy input per day to come from starchy carbohydrates.

For his evening meal he has decided to have pasta, the only carbohydrate he eats in the day. He wants to know the energy content of 1 g of pasta in order to calculate how much pasta to allow himself in his evening meal.

Carwyn and Anwen used the following simple calorimeter to measure the energy content of a 7 g piece of dried pasta.



Examiner
only

The table shows one set of results they obtained.

Type of food	Mass of food (g)	Initial temp of water (°C)	Final temp of water (°C)	Increase in temp (°C)	Energy released per gram of food (J)	Energy released per gram of food (kJ)
Pasta	7	19	65	46		

(a) (i) **Complete the table** above by using the following formula. [2]

Energy released from pasta per gram (J) =
$$\frac{\text{volume of water (cm}^3\text{)} \times \text{temp increase (}^\circ\text{C)} \times 4.2}{\text{mass of pasta sample (g)}}$$

Space for working

(ii) Calculate the mass of pasta Carwyn can eat in his evening meal to get one third of his daily energy intake. [2]

Mass of pasta = g

(iii) Carwyn was surprised to discover that he could have that much pasta for his evening meal. Anwen said ‘there’s something wrong, that’s over ten bags of pasta! We have to evaluate our scientific method to find where the error lies.’

Evaluate the design of the apparatus and identify **one** source of error. [1]

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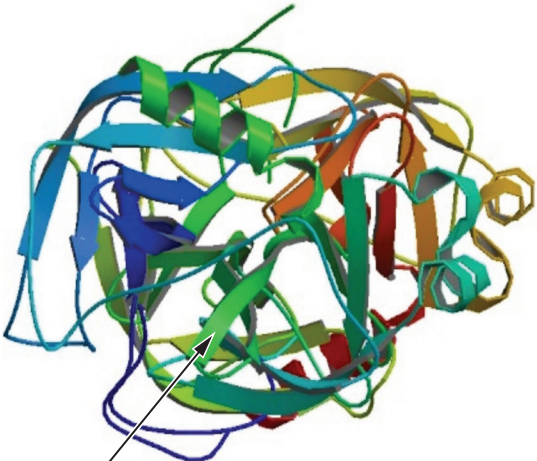
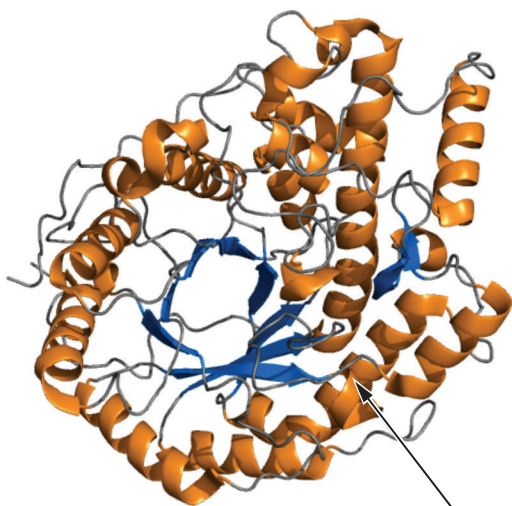
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(b) Carbohydrase is the enzyme that digests the starch in pasta into simple sugars. Protease is an enzyme that digests protein.

Structure of a molecule of carbohydrase

Structure of a molecule of protease



long chains of amino acids joined together

With reference to the diagrams suggest how each of the enzyme molecules are different and state why it is important that the chains of amino acids are folded. [2]

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END OF PAPER

